

6 Digit Nixie Clock "64Bit" - Driver Board

64Bit comes from the fact that this clock is driven with shift registers. Between the 6 nixie digits, and indicators, 64 bits are sent from the microcontroller to the shift registers to create the display output

This is an STM32F446 based project.

The clock time keeping is handled via a Real Time Clock (RTC) communicated with via I2C. The STM32 also has an on-board RTC that can be utilized. Both RTCs have a battery backup via a replaceable 2032 coin cell.

The interface with the clock is via a rotory encoder with a push button. Interfacing with remote capacitive buttons is possible via connectors on the bottom of the Driver Board.

There is a full-color LCD display using driving chip ST7789V2. Communication occurs via SPI.

This clock was designed with safety in mind. There is a dedicated safety circuit that cuts the input and output connections of the high voltage power supply via relays. There are connectors for interfacing external spring pin connections to the cover of the clock, ensuring the high voltage power supply is not on while the lid is off. The safety circuit features dual channels and feedback to the microcontroller. With the spring pin feature, it will provide isolation from the high voltage, even in an event where the microcontroller experiences an error.

The power design of the clock relays on simple components. There are two inputs available to power the clock. A USB-C port and a 12V barrel jack. A boost converter utilizing XL6009 converts (boost) the incoming voltage to 12VDC. Then an LDO creates 5V then another LDO, in series, creates 3.3V. The high voltage power supply typically is set to output 170VDC to 180VDC. It is supplied by a power supply module that sources 12V power.

This driver board interface with the display board PCB via right angle board-to-board connectors. Mounted to the back of the display board is an LED board to provide RGB backlighting to each IN-12 Nixie Tubes.

The circuit boards will be mounted to a 3D printed base via brass standoffs. Overtop of the PCBs will be a glass top. The glass top will have sections where copper tape is applied. Spring pins contact the copper tape to pass connections for the safety circuit.

STM32 MCU

I2C

Interface

Safety Circuit

Power

Connectors

Connectors 2

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Sheet: /

File: 64 Bit - Main Board.kicad_sch

Title: Project Overview

Size: A4

Date: 2025-OCT-04

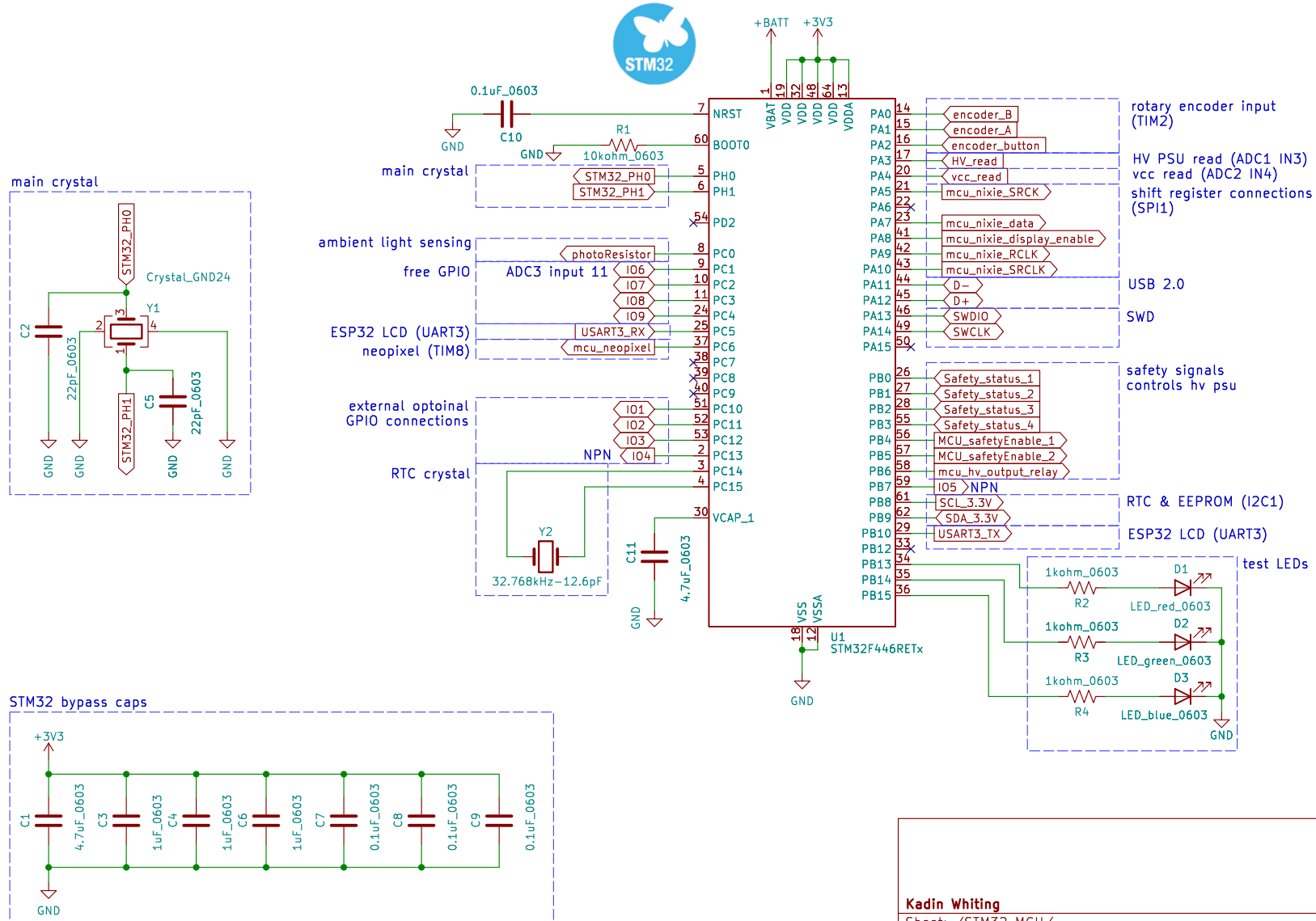
Rev: 1

KiCad E.D.A. 9.0.4

Id: 1/9

STM32F446 Processor ARM M4
168MHz

Level Shifting



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Sheet: /STM32 MCU/

File: STM32 MCU.kicad_sch

Title: Microcontroller

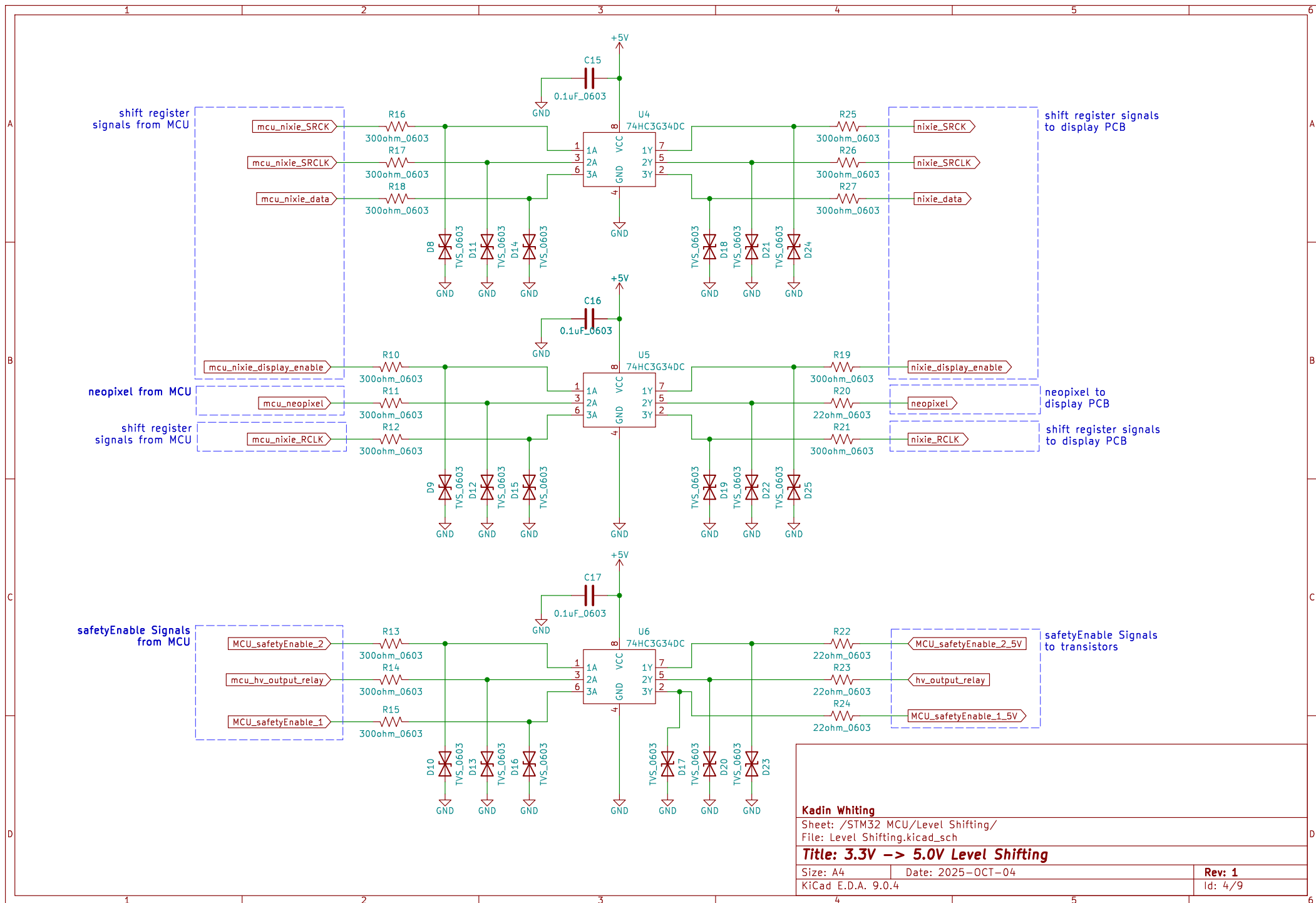
Size: A4

Date: 2025-OCT-04

Rev: 1

KiCad E.D.A. 9.0.4

Id: 2/9



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Sheet: /STM32 MCU/Level Shifting/
File: Level Shifting.kicad_sch

Title: 3.3V -> 5.0V Level Shifting

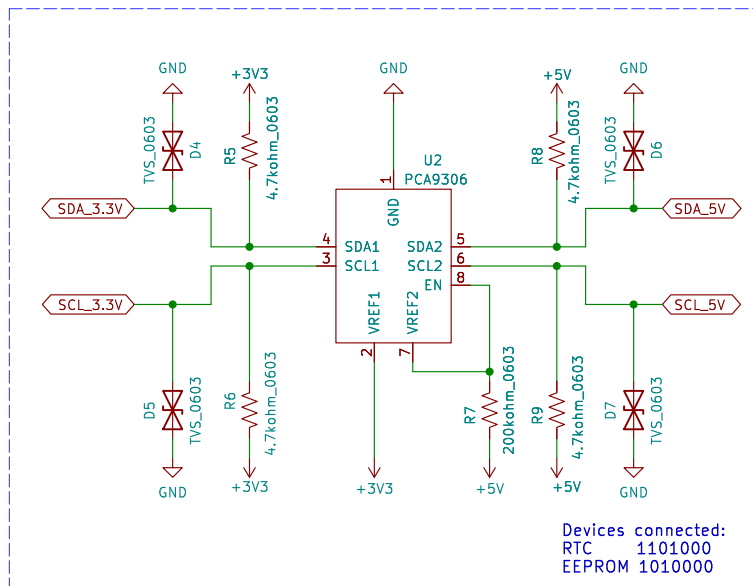
Size: A4 Date: 2025-OCT-04

KiCad E.D.A. 9.0.4

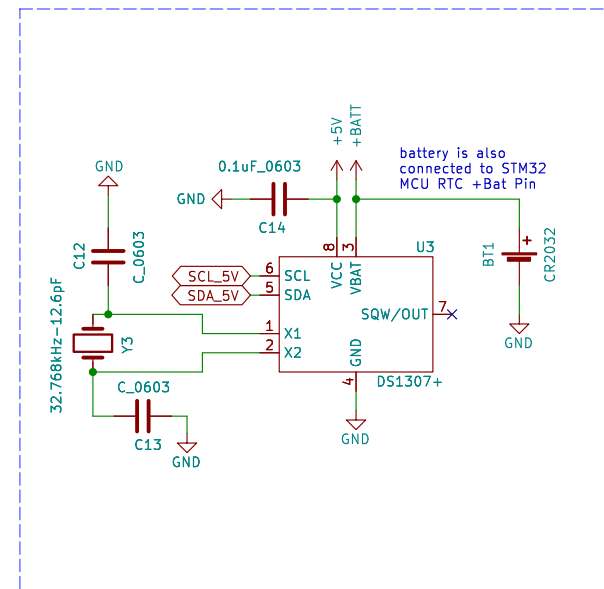
Rev: 1

Id: 4/9

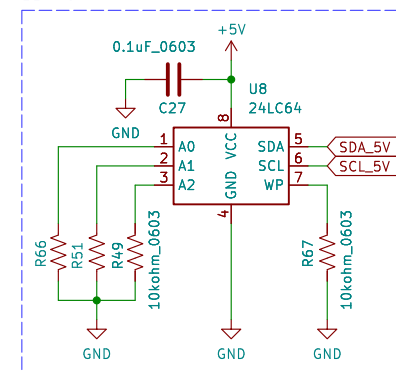
3.3V to 5V I2C level shifter



Real Time Clock



EEPROM



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Sheet: /I2C/
 File: I2C.kicad_sch

Title: RTC & RTC Level Shifting

Size: A4 Date: 2025-OCT-04

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Rev: 1

Id: 3/9

right-angle encoder 24 pulses/rotation
integrated momentary button
low pass filters included

encoder_A
encoder_B
encoder_button

+3V3
+3V3
+3V3

10kohm_0603
4.7kohm_0603
10kohm_0603
10kohm_0603
4.7kohm_0603
10kohm_0603

R29
R30
R31
R32
R33

10kohm_0603
10kohm_0603
10kohm_0603
10kohm_0603
10kohm_0603
10kohm_0603

D28
TVS_0603
C18
D26
TVS_0603
C19
D27
TVS_0603
C20

0.1uF_0603
0.01uF_0603
0.01uF_0603
0.01uF_0603
0.01uF_0603
0.01uF_0603

A1
B1
C1
D1
NO_2
MH1
MH2
NO_1

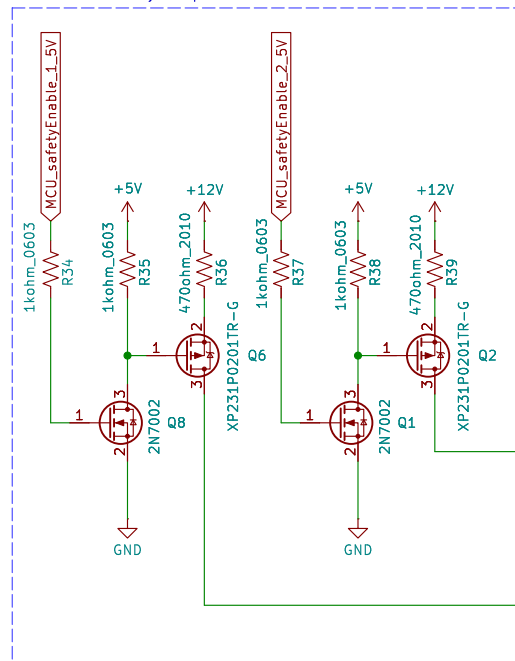
U7
EN12-VS20AF20

E1
E2

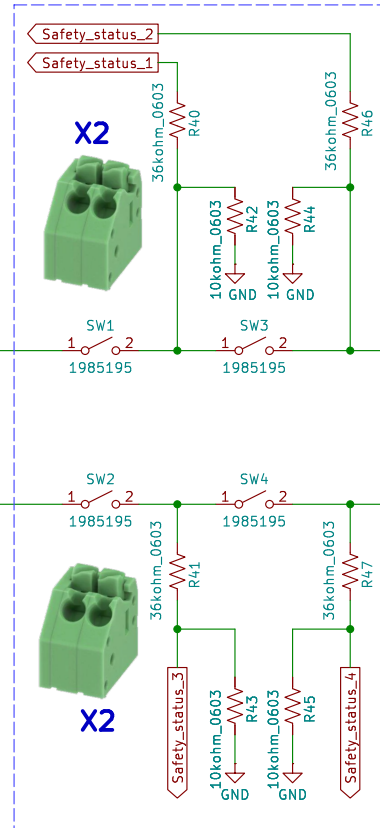
GND
GND
GND
GND
GND
GND
GND
GND
GND

Id: 5/9

two outputs from MCU
to enable safety loop



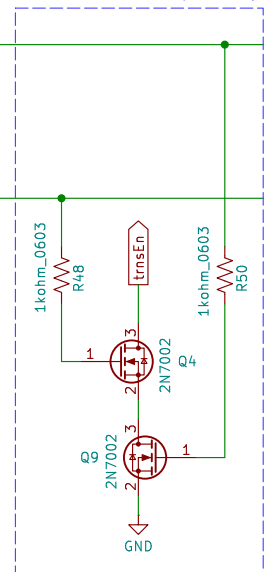
safety contacts for sensing if the lid is installed
feedback to MCU for each contact



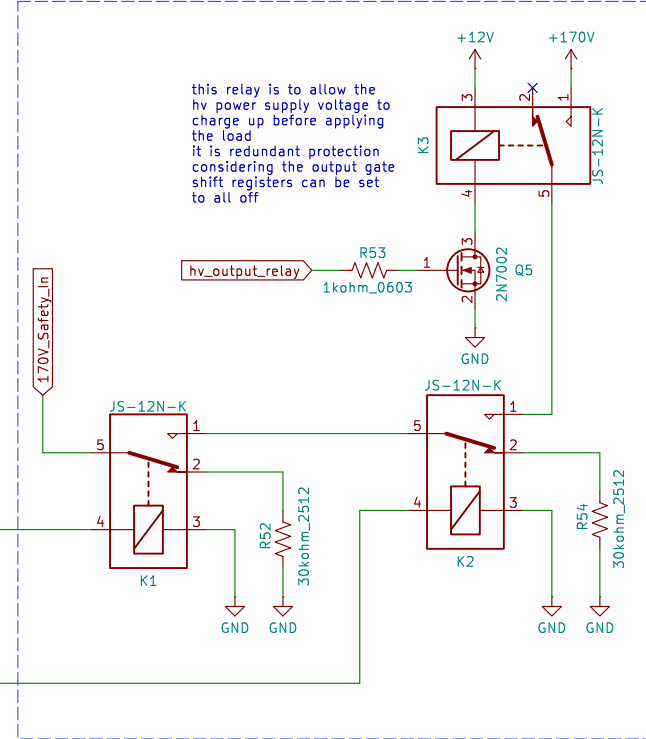
loop through
display PCB



enables HV PSU input relay



controls output of HV power supply



this relay is to allow the
hv power supply voltage to
charge up before applying
the load
it is redundant protection
considering the output gate
shift registers can be set
to all off

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Sheet: /Safety Circuit/

File: Safety Circuit.kicad_sch

Title: Safety Circuit – High Voltage PSU Control

Size: A4

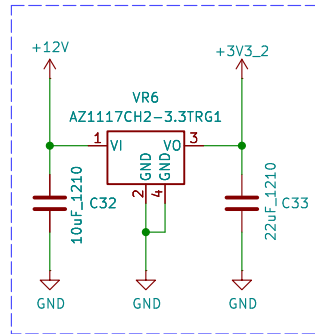
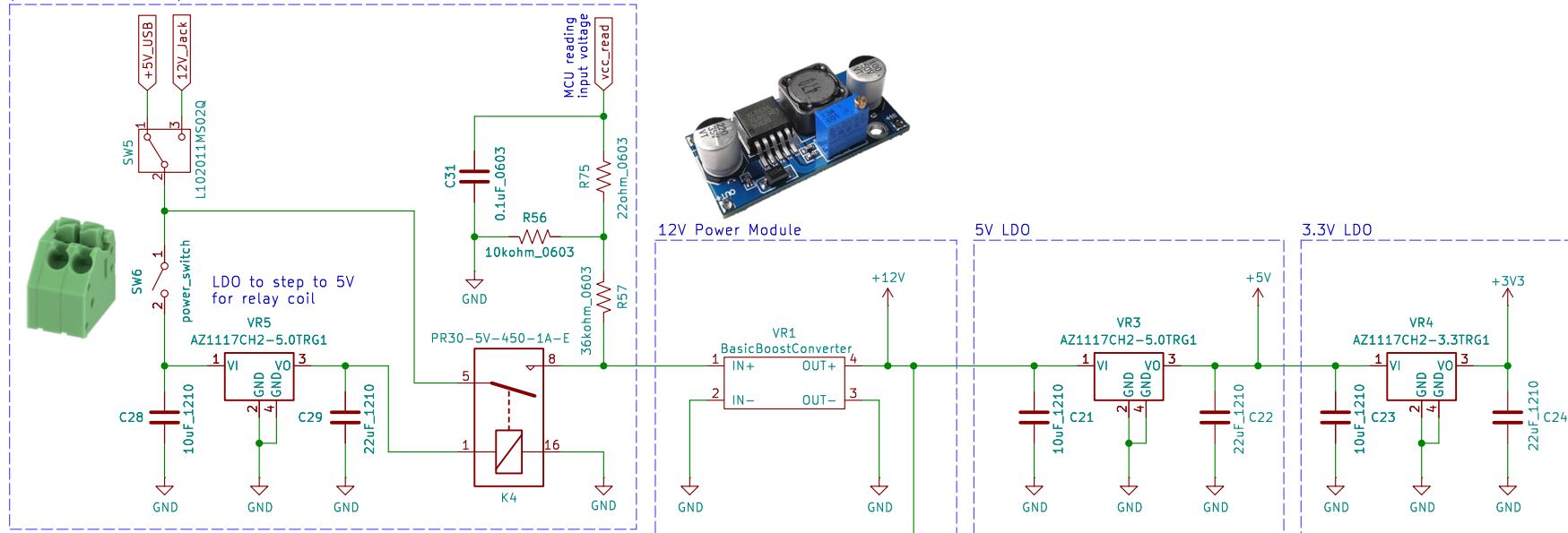
Date: 2025-OCT-04

Rev: 1

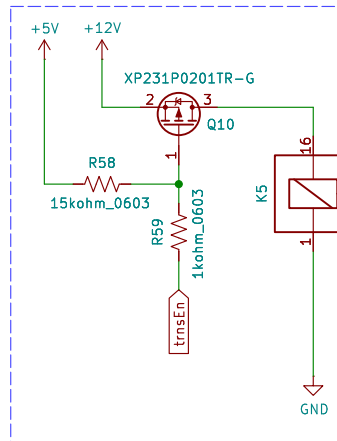
KiCad E.D.A. 9.0.4

Id: 6/9

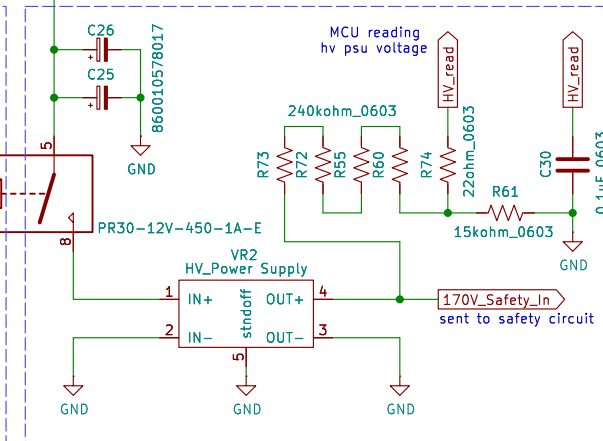
power selection and power switch



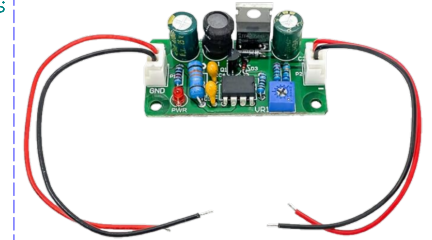
3.3V LDO - external devices



Controlled by safety circuit



high voltage power supply



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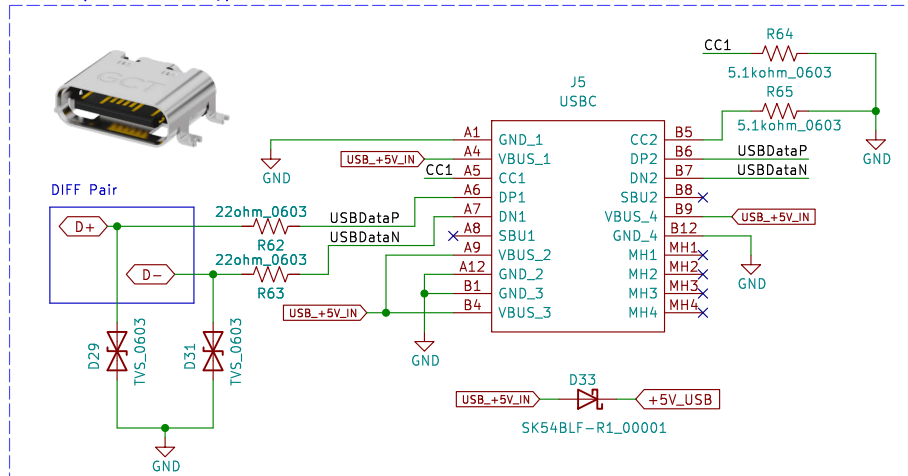
Sheet: /Power/
File: Power.kicad_sch

Title: Power

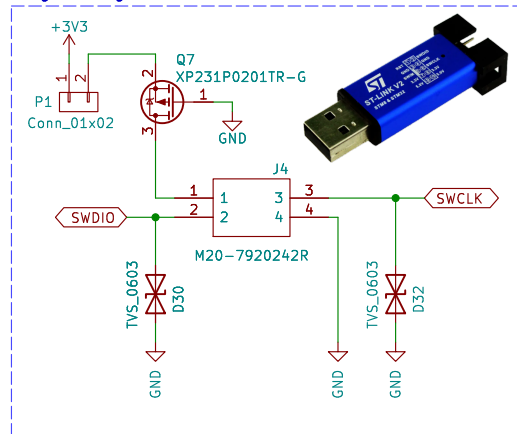
Size: A4 Date: 2025-OCT-04
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Rev: 1
Id: 7/9

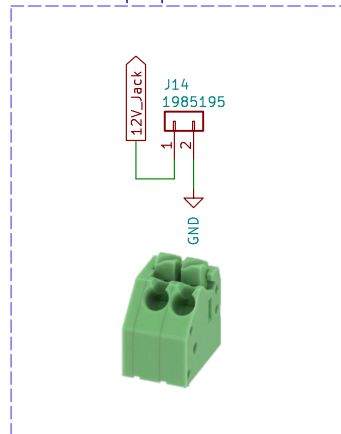
USB-C (2.0 Device Only)



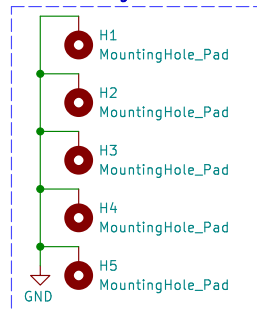
Programming Header



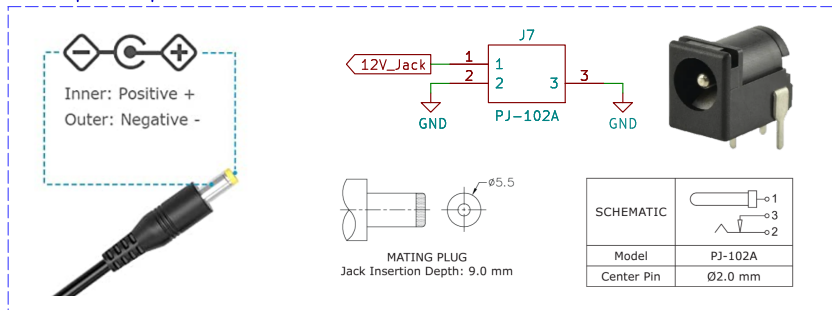
12VDC backup input



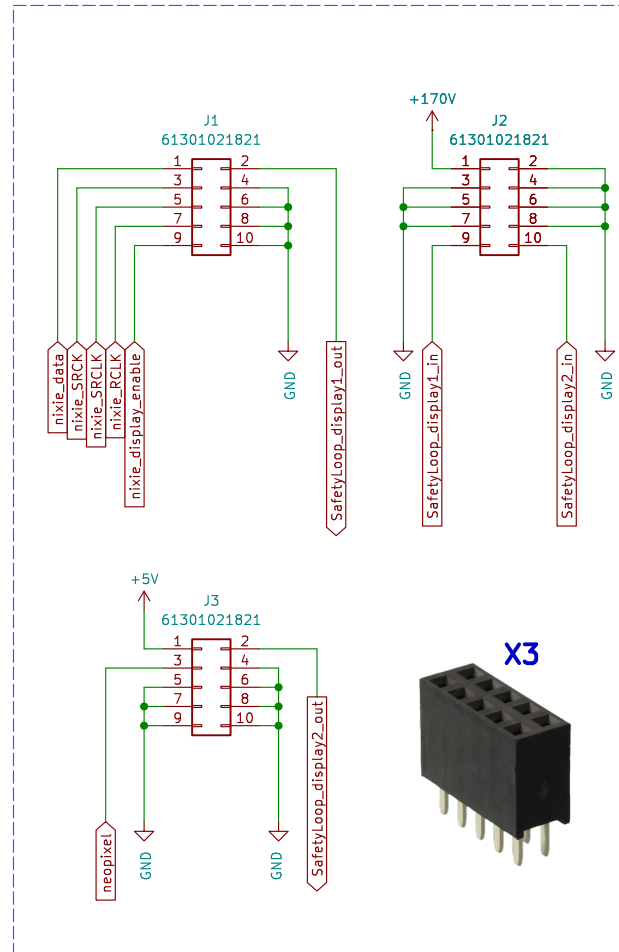
M3 mounting holes



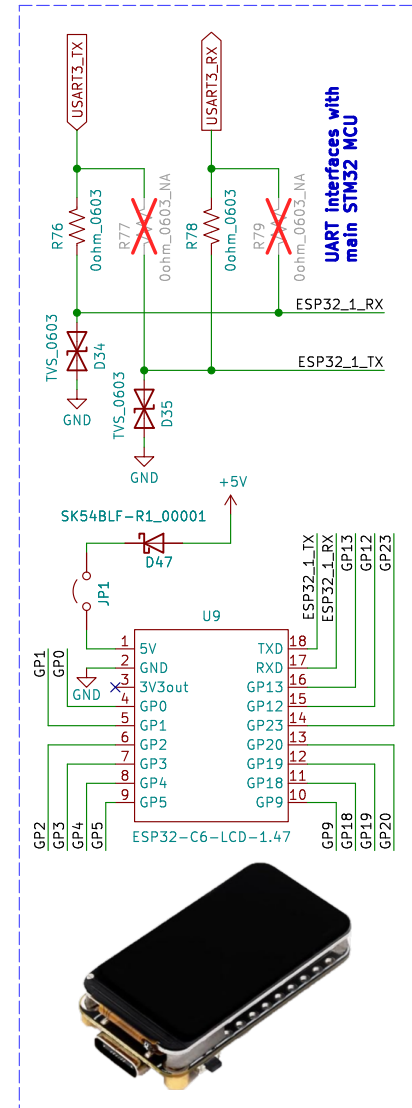
12VDC power input



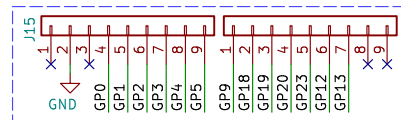
connections to driver board



ESP32-C6 1.47" 172x320 LCD



ESP32 Breakout Connections



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Sheet: /Connectors/

File: Connectors.kicad_sch

Title: Connectors

Size: A4

Date: 2025-OCT-04

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Rev: 1

Id: 8/9

external GPIO optional connections

